



Beata Zalewa

**Creating Custom AI Copilots: Leveraging
Azure OpenAI for Tailored Solutions**



**Expert
Summit 2024**

All about Microsoft® technologies

ONLINE

Sala B

Godz. 10:00

lingaro

SUPREMO®

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TECHNOLOGY
INNOVATION
DATA
KNOWLEDGE **tidk**

About me



Lead Technical Architect



Consultant



Microsoft Certified Trainer



AI & Cybersecurity Practitioner



Developer



Freelancer



Azure @ Heart



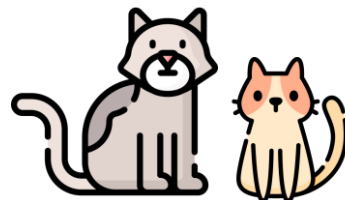
Google Cloud



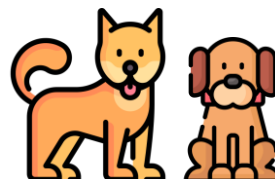
1 Husband



1 Daughter



2 cats



2 dogs



Detective stories



Photography

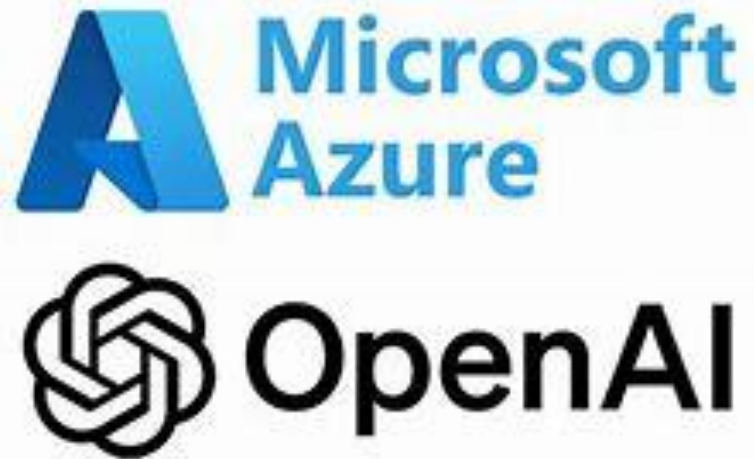
Agenda

- Introduction to Azure OpenAI
- Introduction to Copilots
- Challenges and Considerations
- Q&A Session





Let's go !

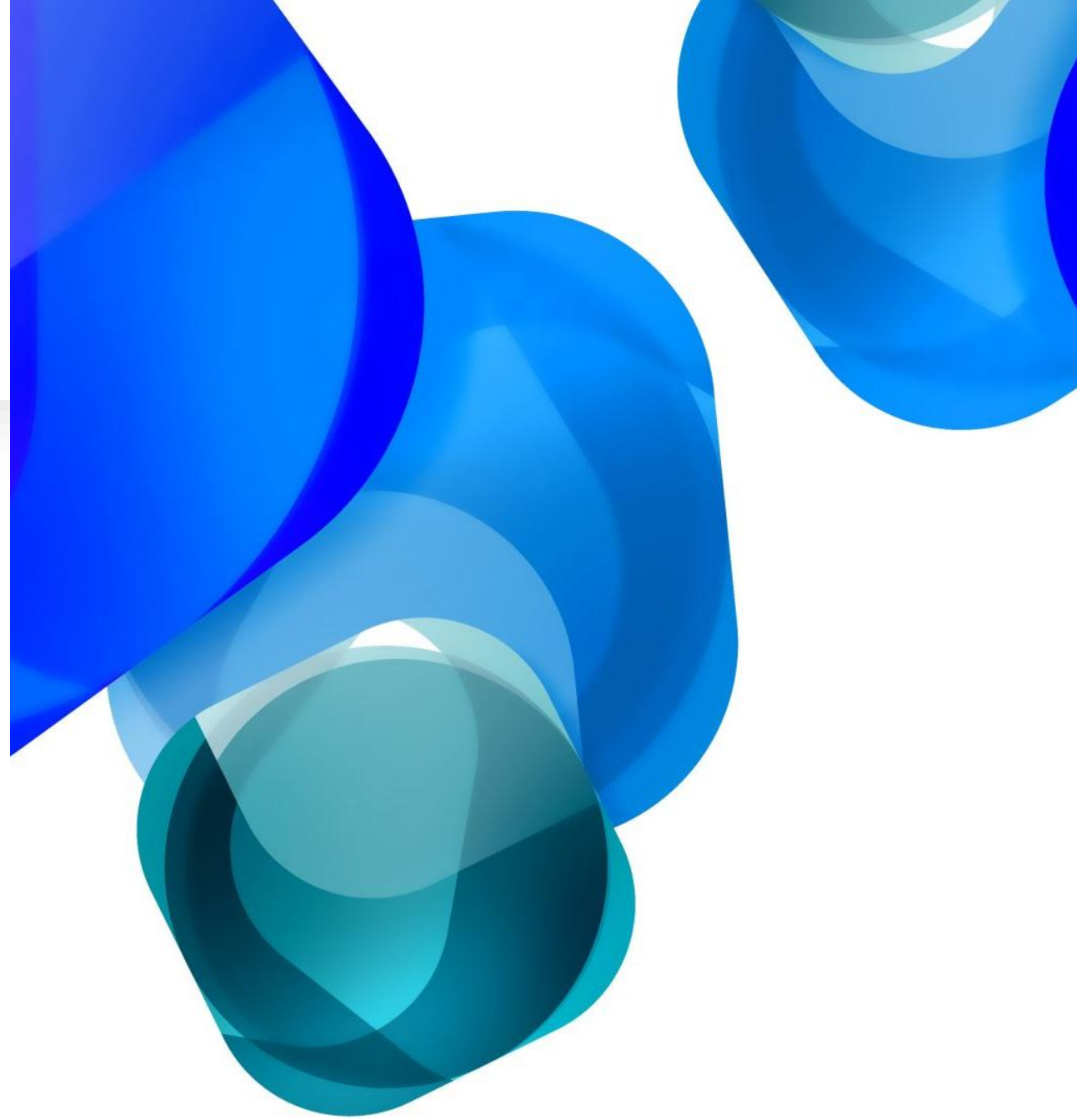


What is Azure OpenAI?

- A cloud-based service providing access to advanced AI models.
- Integrates with Microsoft Azure for scalability and security.
- Supports various AI tasks like language understanding and generation.

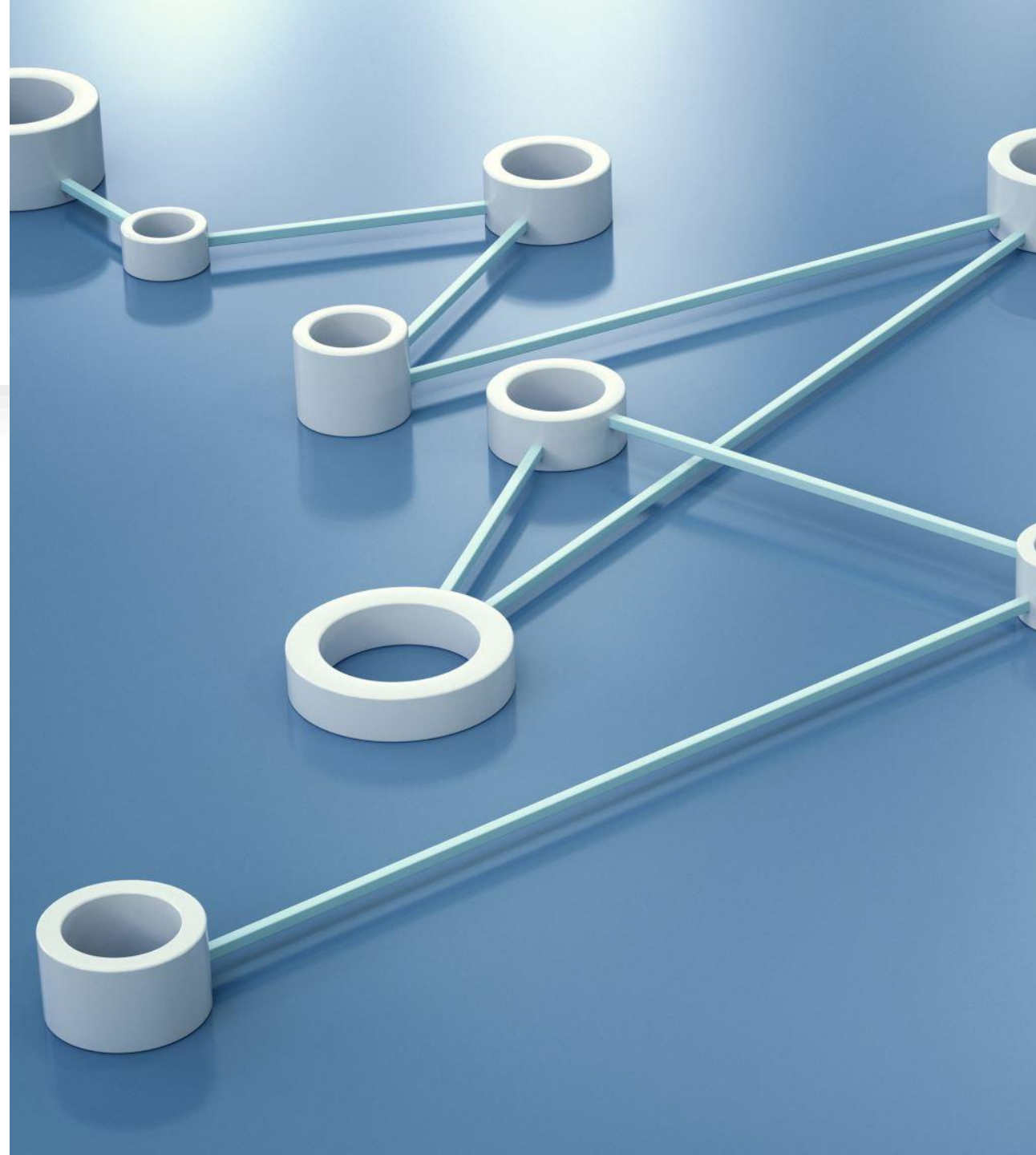
Benefits of Using Azure OpenAI

- Seamless integration with existing Azure services.
- High scalability to handle varying workloads.
- Enhanced security and compliance features.



Key Features of Azure OpenAI

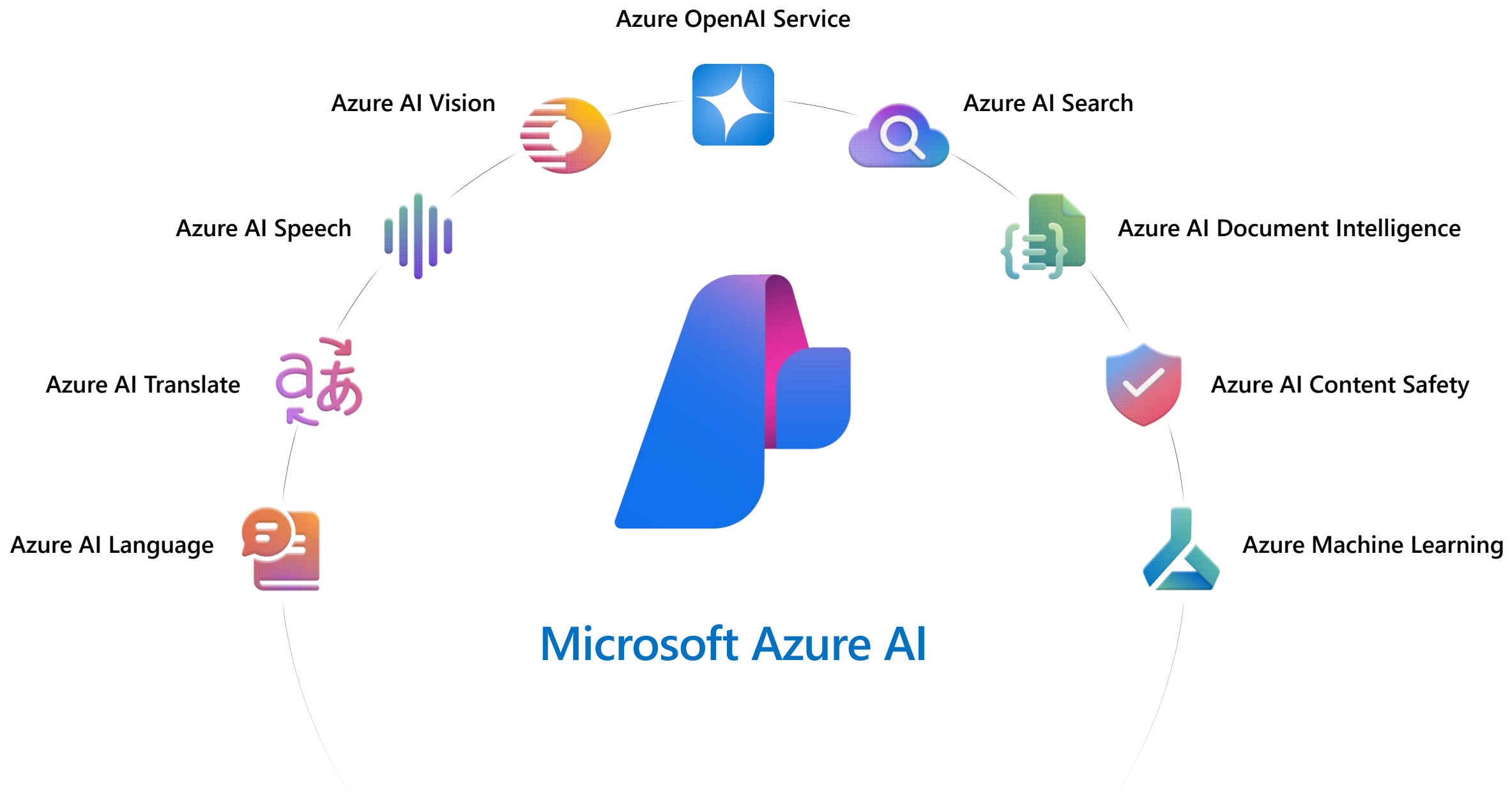
- Access to powerful models like GPT-4 and Codex.
- Fine-tuning capabilities for specific business needs.
- Built-in tools for responsible AI use and content moderation.



Setting Up Your Environment

- Prerequisites for using Azure OpenAI.
- Creating an Azure account and subscription.
- Setting up the Azure OpenAI resource.





Azure AI

AI foundation models



Azure AI Services

Pre-trained solutions for intelligent applications



Azure Machine Learning

Full-lifecycle tools for designing and managing AI models



Responsible AI Tooling

Build and manage apps that are trustworthy by design



Azure AI Studio

A comprehensive platform to develop and deploy custom copilots

Generally available



Azure AI Studio

A comprehensive platform to develop and deploy custom copilots

API and model
choice

Complete AI
toolchain

Responsible AI
tools and practices

Enterprise-grade
production at scale

ai.azure.com

Azure AI Studio

Azure AI Studio

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Model benchmarks

Prompt catalog

Azure OpenAI

AI Services

Management

All resources

Quota

Recent projects

[View all projects](#)

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Project	Hub	Description	Location	Created on
info-demo-expertsummit2024	info-expertsummit-ai-demo		eastus	Nov 7, 2024 8:55 AM
info-demo-expertsummit2024	info-expertsummit-ai-demo		eastus	Nov 7, 2024 8:55 AM

Jump to top tasks and tools

Get started with Azure AI Studio

Learn how you can start developing generative AI applications with next-gen models and tools, via web app or directly in code with the SDK.

[Get started in Azure AI Studio](#)

Assistants API available on Azure OpenAI service

Develop power agent-like experiences with built-in state and thread management, knowledge retrieval, and tools including code interpreter and function calling.

[Build your AI Assistant](#)

Experiment with prompts in the playground

See how different foundation models respond to user input by trying out different prompts, adjusting parameters, and even grounding on your own data.

[Try some prompts in the playground](#)

Deploy large language models (LLMs)

Deploy an LLM or prompt flow and make its API available for use to an application, website, or other production environment.

[View deployments](#)

ai.azure.com

Azure AI Studio

 Azure AI Studio

All resources

info-expertsummit-ai-demo

All projects

info-demo-expertsummit2024

Overview

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Project

info-demo-expertsummit2024







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Project playground

 Chat

 Assistants PREVIEW

 Real-time audio PREVIEW

 Images

 Completions

 Speech PREVIEW

Tools

 Code PREVIEW

 Prompt flow

 Evaluation PREVIEW

 Fine-tuning PREVIEW

info-demo-expertsummit2024 

Pick up where you left off

Once you've deployed a model, connected data, or created any assets, they'll show up here ... so you can always pick up where you left off.

Documentation

[Build and deploy a question and answer copilot with prompt flow in Azure AI Studio](#)

Use generative AI and prompt flow to build, configure, and deploy a copilot for your retail company called Contoso.

[Build a custom chat app in Python using the prompt flow SDK](#)

Develop a custom chat app in your local development environment, and learn to run and evaluate LLM prompts with the Prompt Flow SDK.

[Retrieval augmented generation and indexes](#)

This article talks about the importance and need for Retrieval Augmented Generation (RAG) and Index in generative AI.

[View more](#)

Description 

Describe your project (optional)

Project properties

Hub display name

[info-expertsummit-ai-demo](#) 

Hub resource name

[info-expertsummit-ai-demo](#) 

Subscription

[ZALNET-PAYG](#) 

Resource Group

[ExpertSummitAzureAIDemoRG](#) 

[Manage in Azure Portal](#)

[Get API endpoints and keys](#)

[View subscription quota](#)

Delete project

Project users 2

 Beata Zalewa (You)
info@zalnet.pl

Owner

[View all](#)

ai.azure.com

Azure AI Studio

All projects / info-demo-expertsummit2024 / Deployments / gpt-35-turbo

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gpt-35-turbo

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Open in playground

Edit

Delete

Deployment info

Name gpt-35-turbo	Provisioning state Succeeded
Deployment type Standard	Created on 2024-11-07T08:02:26.3608594Z
Created by 7d923f60-74b7-4b4a-8f49-3e8f1b0eaeaa	Modified on Nov 7, 2024 9:02 AM
Modified by 7d923f60-74b7-4b4a-8f49-3e8f1b0eaeaa	Version update policy Once a new default version is available
Rate limit (Tokens per minute) 120,000	Rate limit (Requests per minute) 720
Model name gpt-35-turbo	Model version 0301
Life cycle status GenerallyAvailable	Date created Mar 9, 2023 1:00 AM
Date updated Jul 6, 2023 2:00 AM	Model retirement date Jan 27, 2025 1:00 AM

Endpoint

Target URI

https://ai-infoexpertsummitaidemo659424480964.openai.azure.com/openai/deployments/...

Authentication type

Key

Key

.....

Monitoring & safety

Content filter

DefaultV2

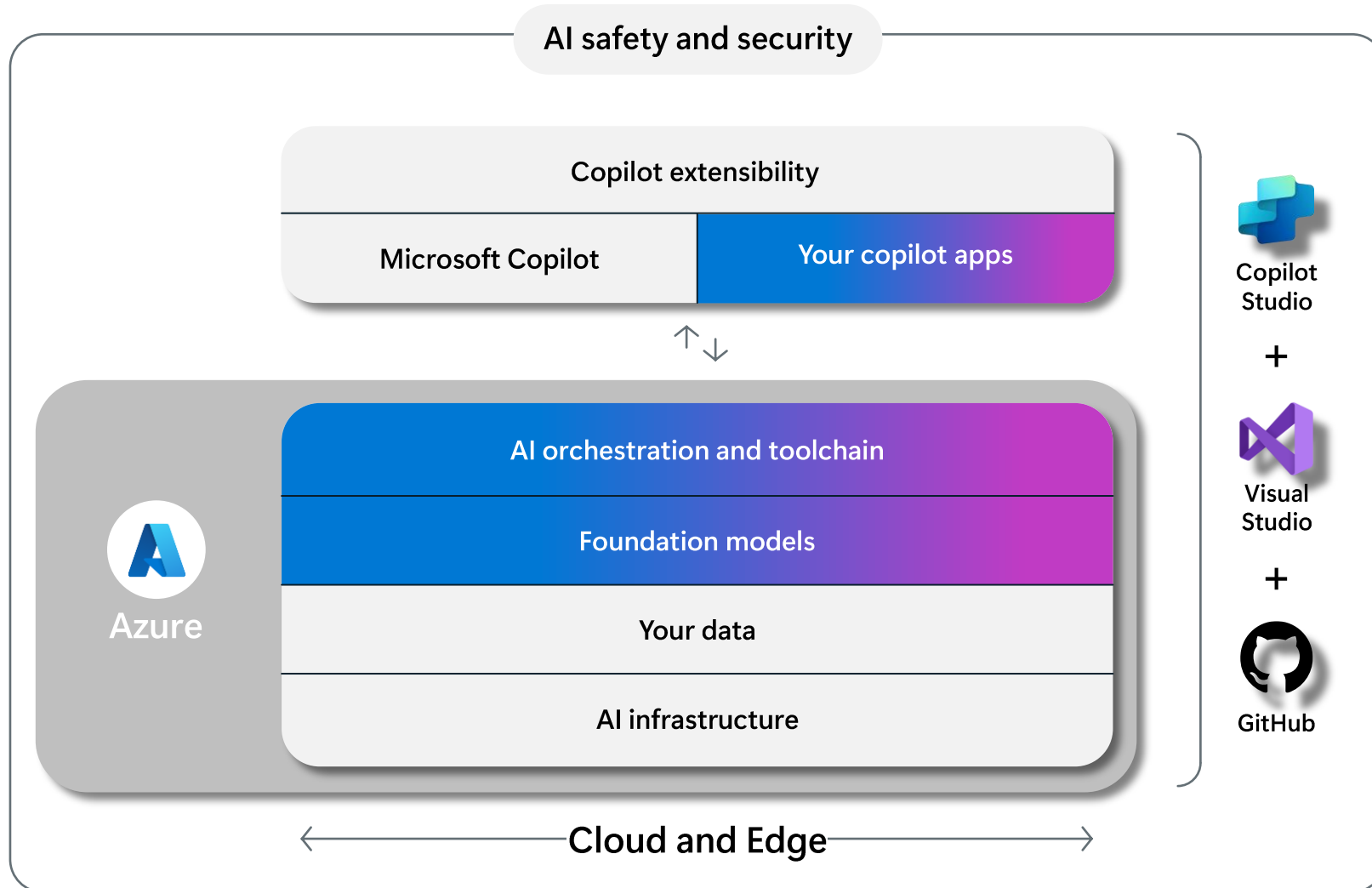
Useful links for application development

[Code sample repository](#)

[Tutorial](#)

ai.azure.com

Copilot stack



Copilot Studio

Azure AI Studio

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All resources & projects

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info-demo-expertsummit2024

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Authentication type

Key

Key

Monitoring & safety

Content filter

DefaultV2

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[Code sample repository](#)

[Tutorial](#)

copilotstudio.microsoft.com

Copilot Studio

The screenshot displays the Copilot Studio web interface. At the top, a navigation bar includes the Copilot Studio logo, a user profile icon, and the environment name 'ZALNET Beata Zalewa'. A sidebar on the left contains icons for Home, Create, Agents, Library, and a menu. The main content area features a large section titled 'Describe your agent to create it' with buttons for 'Helpdesk', 'Expense tracking', and 'HR and benefits'. Below these is a text input field with the placeholder 'Use everyday words to describe what your agent should do' and a submit arrow. A note indicates that this AI-powered feature is in preview. The 'Start with a template' section follows, with a 'See more' link. It contains six template cards: 'Safe Travels' (travel guidelines), 'Website Q&A' (instant answers), 'Team Navigator' (finding colleagues), 'IT Helpdesk' (support tickets), 'Store Operations' (retail efficiency), and 'Case Management' (customer support). The 'Learning resources' section at the bottom lists links for quick starts, documentation, security, responsible AI, support community, release plans, and release notes.

Copilot Studio

Environment: ZALNET Beata Zalewa (...)

Describe your agent to create it

Helpdesk Expense tracking HR and benefits

Use everyday words to describe what your agent should do

This AI-powered feature above is currently in preview. [See terms](#) and [supplemental terms](#)

Start with a template

Templates are currently a preview feature. See [supplemental terms](#)

[See more](#)

- Safe Travels**
Provides answers to common travel questions and related health and safety guidelines
- Website Q&A**
Instantly answer user questions using the content of your web site or other knowledge.
- Team Navigator**
Assists employees in finding colleagues and their hierarchy within the organization using Microsoft Teams
- IT Helpdesk**
Empowers employees to resolve issues and effortlessly create/view support tickets.
- Store Operations**
Improve the efficiency of retail frontline workers by enabling easy access to store procedures and policies
- Case Management**
Provide round-the-clock automated support to customers by understanding their issues and creating cases

Learning resources

- Quick start: Create and deploy an agent
- Documentation
- Security and Governance
- Responsible AI FAQs
- Quick start: Use Generative AI in an agent
- Support community
- Coming soon: See release plans
- What's new: See release notes

Copilot Studio

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Copilot Studio

Environment
ZALNET Beata Zalewa (...)

⚙️ ? BZ

Home

Create

Agents

Library

...

Agents

+ New agent

Import agent

🔍 Search agents

Name	Type	Last modified ↓	Last published	Owner
 Store Operations	Agent	# Microsoft Copilot Studio 4 hours ago	Never	Beata Zalewa
 Safe Travels Copilot 1	Agent	# Microsoft Copilot Studio 4 hours ago	Never	Beata Zalewa
 Weather Copilot 1	Agent	# Microsoft Copilot Studio 4 hours ago	Never	Beata Zalewa
 Contoso Customer Service Demo	Agent	# Microsoft Copilot Studio 5 hours ago	Never	Beata Zalewa
 Website Q&A Copilot	Agent	# Microsoft Copilot Studio 1 month ago	Never	Beata Zalewa
 Copilot demo	Agent	# Microsoft Copilot Studio 5 months ...	5 months ago	Beata Zalewa
 ServiceNow Ticket Status	Agent	Beata Zalewa 6 months ago	Never	Beata Zalewa
 PVA 1 OpenAI Chatbot (classic)	Chatbot (classic)	Beata Zalewa 1 year ago	Never	Beata Zalewa

Copilot Studio

The screenshot displays the Copilot Studio web interface. At the top, a blue header bar contains the 'Copilot Studio' logo, a navigation menu with icons for Home, Create, Agents, and Library, and a right-hand section showing the current environment 'ZALNET Beata Zalewa (...)' with settings, help, and user profile icons. Below the header, the main content area is titled 'Weather Copilot 1' and includes tabs for Overview, Knowledge, Topics, Actions, Analytics, and Channels. On the right side of this section are buttons for 'Publish', 'Settings', and 'Test'. A large blue banner at the top of the main area reads 'Your agent is ready! Here's what's next:' and lists three steps: 'Add actions', 'Build topics', and 'Publish your agent'. The main configuration panel is divided into sections: 'Details' (with an 'Edit' button), 'Description', 'Instructions', 'Knowledge' (with an 'Add knowledge' button and a toggle for 'Allow the AI to use its own general knowledge' which is 'Enabled'), and 'Actions' (with an 'Add action' button). The 'Actions' section lists two actions: 'MSN Weather - Get current weather' and 'MSN Weather - Get forecast for today'. On the far right, a 'Test your agent' section is visible, showing a sample chat interaction where the agent introduces itself as 'Weather Copilot 1'.

Weather Copilot 1 Overview Knowledge Topics Actions Analytics Channels

Your agent is ready! Here's what's next:

- ⚡ [Add actions](#) so your agent can do things for you
- 🗂️ [Build topics](#) to focus and guide how your agent answers
- 👤 [Publish your agent](#) so others can use it

Details [Edit](#)

Name
Weather Copilot 1

Description
Your go-to assistant for getting weather forecast

Instructions
You should remain friendly and polite at all times. Do not answer questions that are not related to the weather.

Knowledge [+ Add knowledge](#)

Add data, files, and other resources to inform and improve AI-generated responses.

Allow the AI to use its own general knowledge. [Learn more](#) ☒ **Enabled**

Actions [+ Add action](#)

Add actions to empower the AI to complete specific tasks for improved engagement.

- ☀️ MSN Weather - Get current weather ...
- ☀️ MSN Weather - Get forecast for today ...

Test your agent ... ↻ ✕

Hello, I'm Weather Copilot 1, a virtual assistant. I can answer general questions about the current weather as well as forecasts for today and tomorrow. Just so you are aware, I sometimes use AI to answer your questions.

Note: You can now customize this copilot's topics and knowledge for your own needs. For more information about this template and how to modify it, visit the template [documentation](#).

Use the option that **fits** your scenario **the best**



Azure AI Studio

Offers a wide range of **AI services and models** to build the entire AI development lifecycle

Fast and frictionless **prototyping for quick results**

Azure AI Studio offers **state of the art evaluation, safety** and testing tools for your LLM application



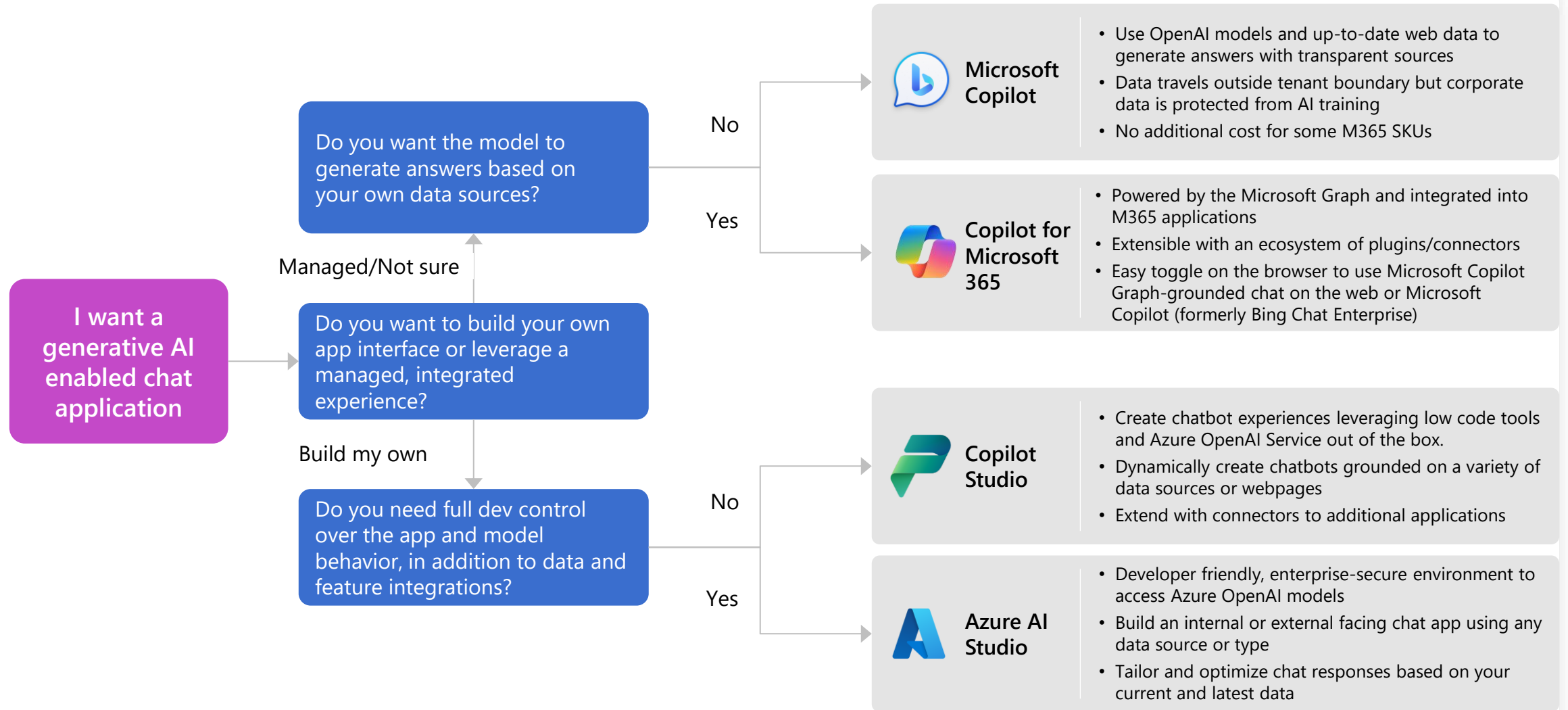
Copilot Studio

1400+ connectors available out-of-the-box + you can create your own connector

Channels like **Microsoft Teams, Facebook, and Slack** are easy to configure

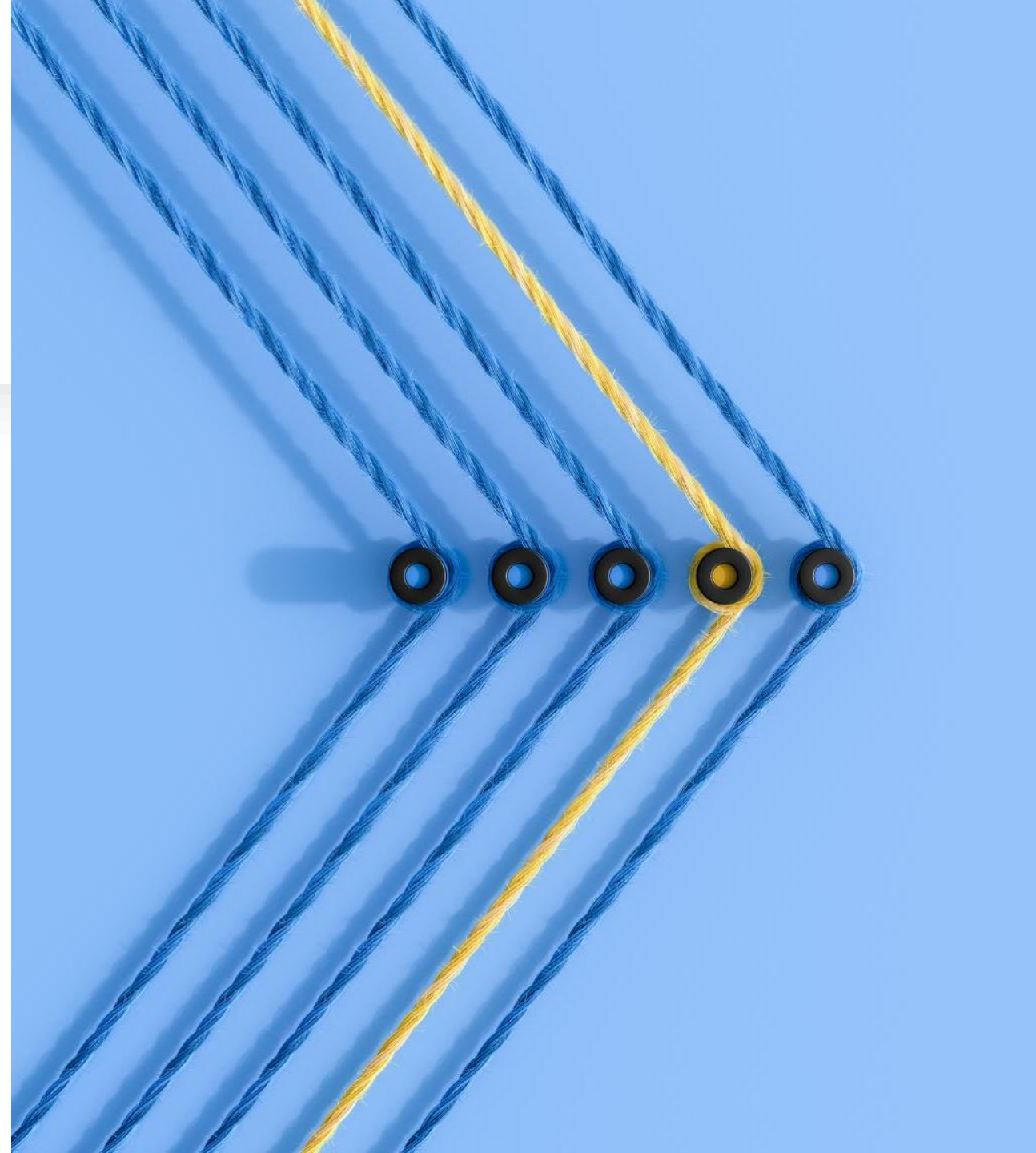
Enterprise dialog management and conversational orchestration

Microsoft AI Decision Guide (Buy or Build)



Building Your First AI Copilot

- Step-by-step guide to creating a basic project.
- Using the Azure OpenAI Studio for model selection.
- Deploying your first model with sample data.



Integrating with Existing Systems

- Connect Azure OpenAI with other Azure services.
- Examples of integrations (e.g., Azure Functions, Logic Apps).
- Ensuring smooth data flow between systems.



Security Considerations

- Overview of security features in Azure OpenAI.
- Best practices for securing your AI applications.
- Compliance with data protection regulations.



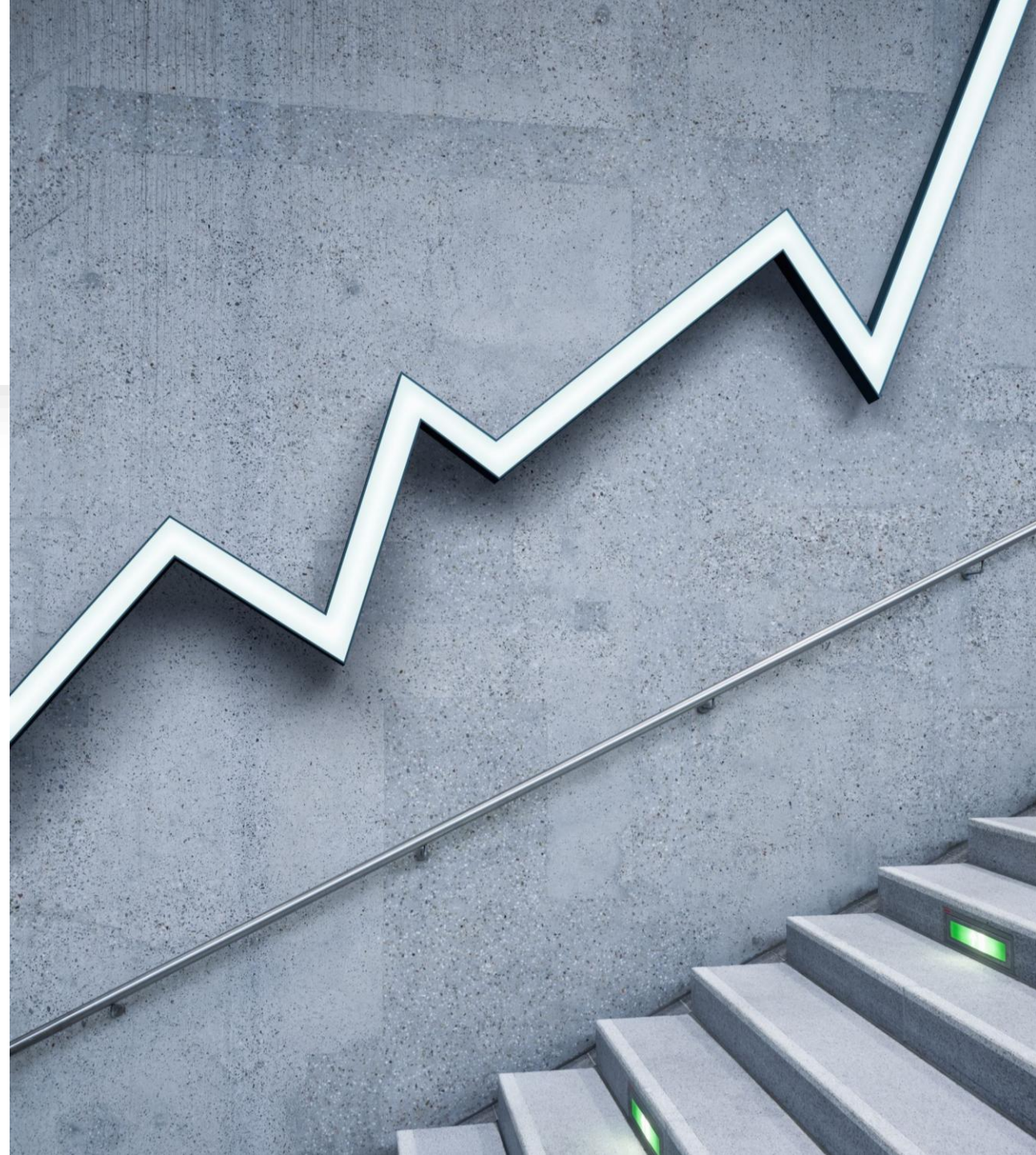


Building a Team for Success

- Skills required for developing custom AI solutions.
- Importance of cross-functional collaboration in projects.
- Encouraging a culture of innovation within teams.

Measuring Success

- Key performance indicators (KPIs) to track effectiveness.
- Gathering user feedback to assess impact and satisfaction.
- Iterative improvement based on performance metrics.



Ethical Considerations in AI Development

- Azure OpenAI emphasizes ethical AI use with built-in content filtering and monitoring features.
- Ensures compliance with privacy regulations and responsible data handling practices.
- Supports diversity and inclusion initiatives in AI development..



Cost-Effectiveness

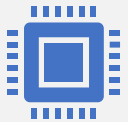
- Pay-as-you-go pricing model allows businesses to pay only for what they use.
- Flexible plans available for startups, enterprises, and non-profits.
- Cost savings from reduced infrastructure management and resource allocation.



Pre-Trained Models



Access to various pre-trained models for tasks like text summarization and sentiment analysis.



Fine-tune models with minimal coding to meet specific business needs.



Reduces time and resources needed for model training from scratch.

Custom Model Creation



Use OpenAI Codex to create custom models tailored to specific applications.



Capable of generating code, text, images, and more from natural language inputs.



Facilitates rapid development of applications across various domains.

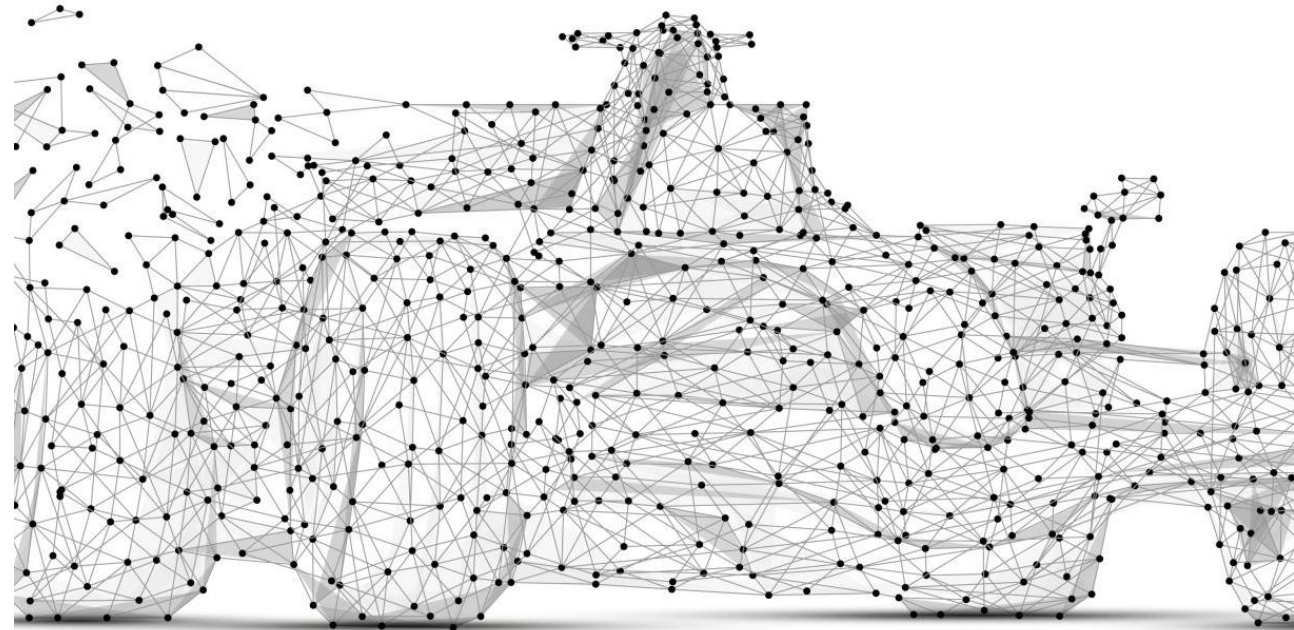
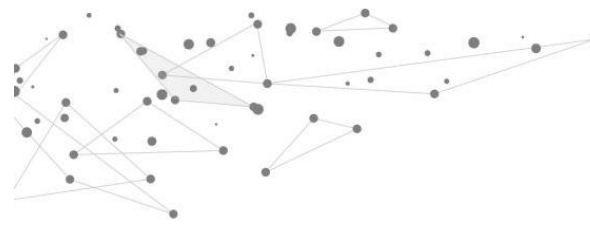


Enhanced Collaboration

- Integration with tools like Azure Machine Learning and Visual Studio Code.
- Web-based OpenAI Playground allows teams to experiment and share results easily.
- Promotes collaboration among developers, data scientists, and business stakeholders.

Use Cases in Business

- Automate customer service with intelligent chatbots powered by Azure OpenAI.
- Generate content for marketing, product descriptions, and internal documentation.
- Analyze large datasets to gain insights and drive decision-making processes.



Improved Customer Experience

- Use AI to enhance user interactions through personalized recommendations and support.
- Streamline processes like resume screening or document summarization for HR departments.
- Create engaging conversational agents that improve customer satisfaction.



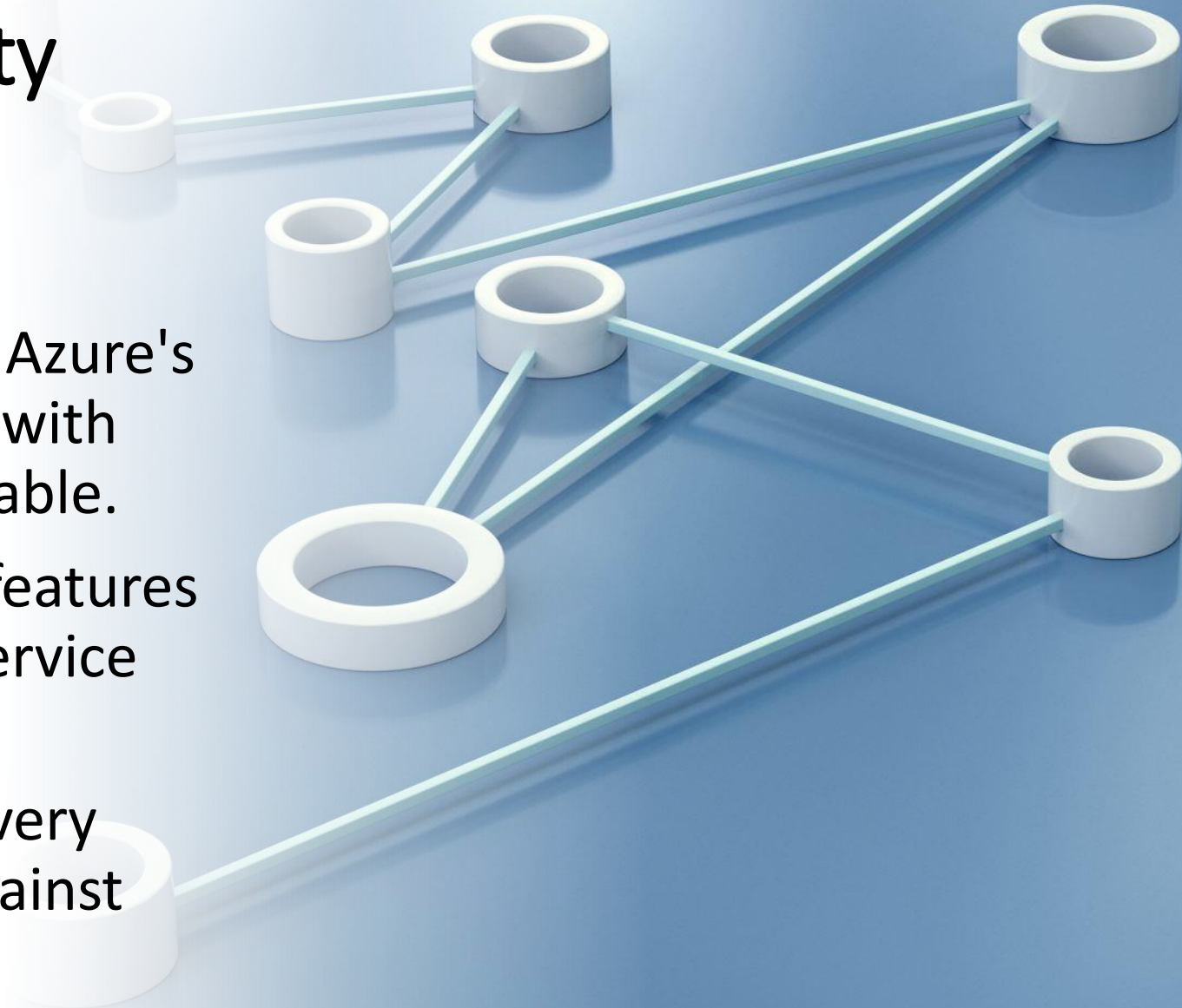


Integration with Microsoft Ecosystem

- Leverage existing Microsoft tools such as Power BI, Dynamics 365, and Teams.
- Enhance productivity by integrating AI capabilities into familiar workflows.
- Benefit from a unified platform that supports various business functions.

High Availability and Reliability

- Hosted on Microsoft Azure's global infrastructure with over 60 regions available.
- Built-in redundancy features ensure continuous service availability.
- Robust disaster recovery options safeguard against data loss.



Training AI Models Efficiently

- Azure provides the necessary computing power for training complex AI models.
- Access to vast datasets enhances the quality of trained models.
- Accelerates the development cycle from ideation to deployment.



Community Support and Resources

- Extensive documentation, tutorials, and code samples available for developers.
- Active community forums provide a platform for sharing experiences and solutions.
- Continuous updates ensure users stay informed about new features and best practices.



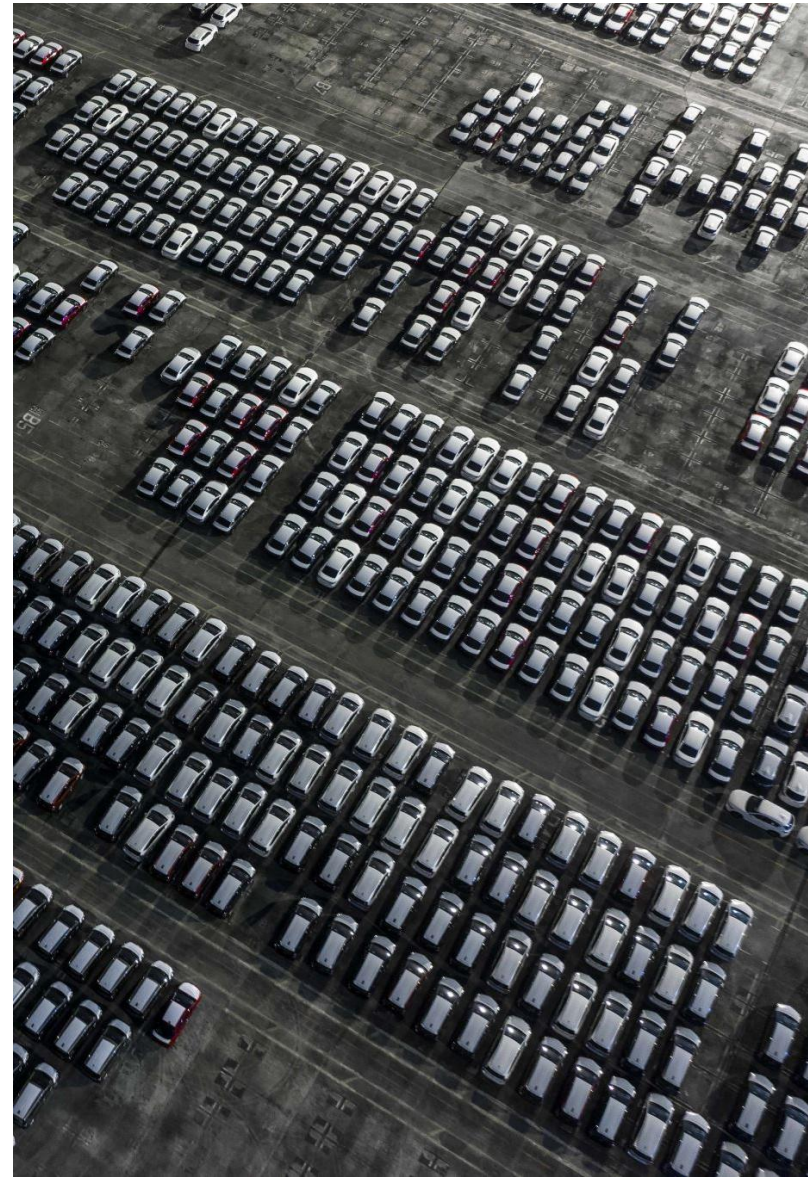
Challenges of Implementation

- Initial learning curve for teams unfamiliar with AI technologies.
- Ensuring data privacy while utilizing third-party AI services can be complex.
- Ongoing maintenance and updates required to keep models effective.



Real-Life Success Stories

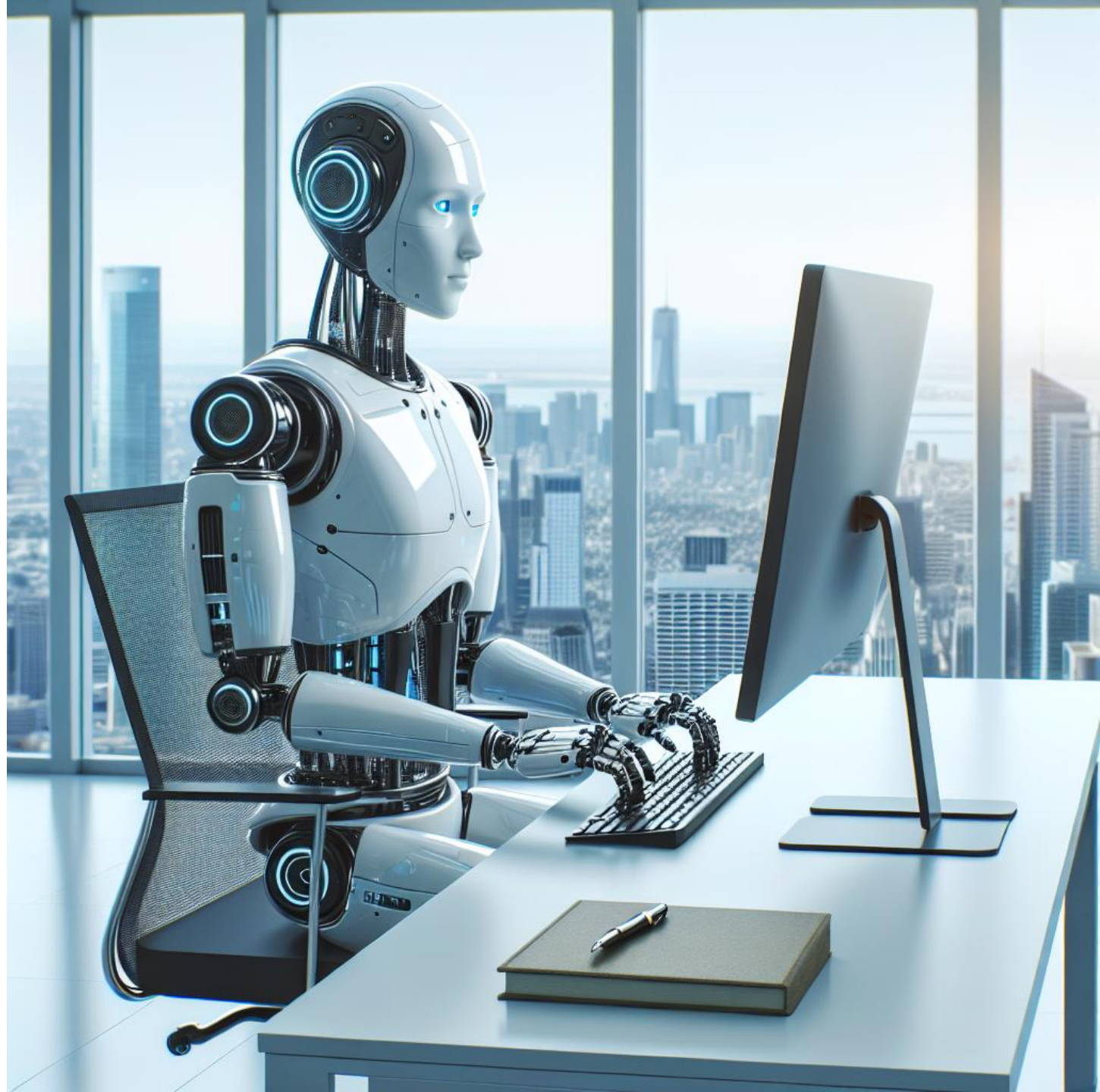
- Companies like CarMax use Azure OpenAI to streamline content creation processes significantly.
- TomTom leverages AI for immersive in-car infotainment systems, enhancing user experience.
- Fraport integrates chatbots to assist employees in operational tasks efficiently.



Training Resources Available

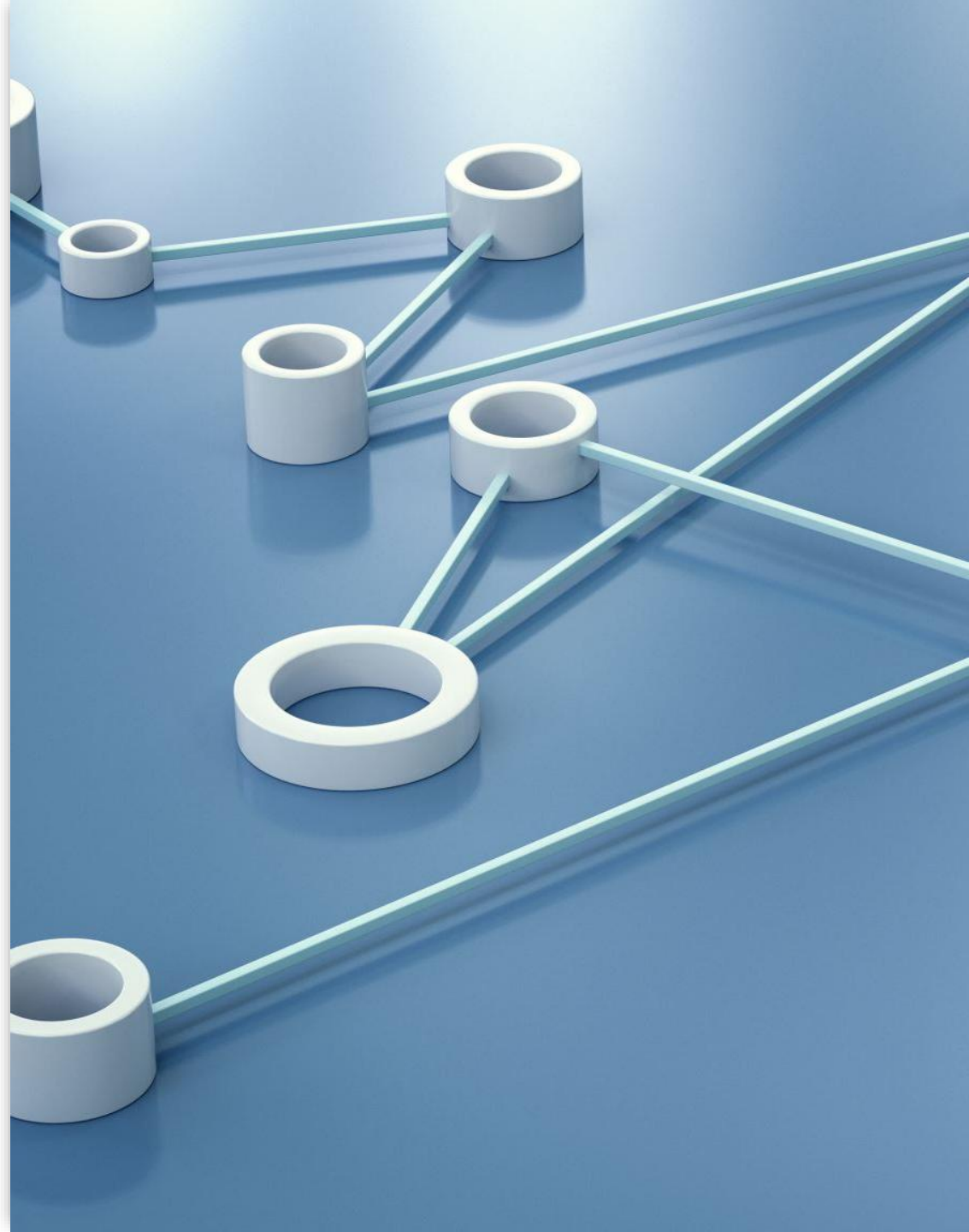
- Explore Microsoft Learn for comprehensive training modules on Azure OpenAI Service.
- Participate in community challenges or hackathons focused on AI development.
- Utilize online courses that cover best practices in implementing AI solutions.

Demo



Conclusion

- Azure OpenAI offers a powerful platform for businesses looking to harness the potential of AI technology.
- Combining scalability, security, and integration capabilities makes it an attractive option for enterprises of all sizes.
- Embrace the future of AI by leveraging Azure OpenAI solutions tailored to your business needs.



I am actively seeking new opportunities and exciting challenges. If you would like to get in touch, please feel free to reach out through the following channels:

Email: info@zalnet.pl

LinkedIn: <https://www.linkedin.com/in/beatazalewa/>

Blog: <https://beatazalewa.com/>

Twitter: <https://x.com/beatazalewa>

GitHub: <https://github.com/beatazalewa/>



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